SPACE X
LAUNCH SUCCESS ANALYSIS

IBM Applied Data Science Capstone Project

Data collection • Data wrangling • EDA • SQL • Interactive visualization • Predictive analytics

Student: _____
GitHub repository: ADD YOUR REPOSITORY URL

Emergency submission draft. This document provides a project structure but does not claim that the analyses or models have been executed.

1. Executive Summary

Project overview

This project studies factors associated with Falcon 9 first-stage landing outcomes. It brings together public launch data, web scraping, data cleaning, exploratory data analysis, SQL summaries, geospatial mapping, an interactive dashboard, and supervised classification. The practical question is whether observable mission characteristics can help estimate the likelihood of a successful first-stage landing.

- Collect launch, site, payload, and outcome records from the course sources.
- Clean missing values, dates, categories, and identifiers; define a consistent binary target.
- Explore relationships among launch site, payload mass, orbit, flight number, and outcome.
- Compare classification models with held-out evaluation and appropriate metrics.

No empirical results are fabricated in this draft. Insert actual counts, charts, model metrics, and screenshots from the course labs before final submission if possible.

2. Introduction and Problem Statement

Background: Falcon 9 reusability is an important part of launch economics. Historical launch records can be analyzed to understand which observed characteristics are associated with landing outcomes.

Problem statement: Can launch characteristics help estimate whether the first stage will land successfully?

- Target: first-stage landing outcome (successful / unsuccessful).
- Potential features: launch site, payload mass, orbit, flight number, booster version, and mission context.
- One row represents one launch; document the exact dataset and target definition.
- Prevent data leakage by splitting data before fitting preprocessing and models.

3. Data Collection — SpaceX API

Use the API exercise to obtain structured launch records. Select relevant fields such as launch identifiers, dates, sites, payloads, booster details, and outcomes. Document the exact endpoint and fields used.

Workflow: Request API → parse JSON → select and normalize fields → validate records → save a tabular dataset.

- Record the endpoint and retrieval date.
- Inspect nested fields and missing values.
- Normalize dates and numeric fields.
- Save the extraction script/notebook and add its GitHub URL.

4. Data Collection — Web Scraping

Use the course-provided web-scraping source to obtain launch tables or supplemental records. Inspect the HTML table, parse rows into a DataFrame, and validate the output.

Workflow: Open source page → inspect table → parse records → validate headers, types, and duplicates.

- Record the exact source URL.
- Check row counts and column names after extraction.
- Compare overlapping fields with API records where possible.
- Add the scraper script and GitHub URL.

5. Data Wrangling Methodology

Task	Method	Validation
Missing values	Inspect missingness; impute or exclude only with documented reasons.	Missing-value summary
Duplicates	Check unique launch IDs and remove confirmed duplicates.	Unique ID check
Data types	Parse dates; convert payload mass and flight number to numeric.	Type/range checks
Target label	Define landing outcome consistently as binary target.	Class distribution
Encoding	Encode categorical variables inside the model pipeline.	Train/test consistency

Add actual before/after row counts and the final feature list from the completed lab.

6. EDA with Visualization

EDA should describe the dataset and explore relationships between launch characteristics and landing outcomes. The following are required plot types, not measured findings.

- Scatter plot: payload mass versus outcome, with color for success/failure.
- Bar chart: launches or successful landings by launch site.
- Yearly trend: launch volume and success rate by year.
- Histogram or box plot: payload mass distribution, optionally grouped by outcome.
- For each chart, state the question answered, units, sample size, and interpretation.

Insert actual charts and data-supported interpretations from the notebook. Do not claim a trend until it has been calculated.

7. EDA with SQL

Use SQL to summarize launch records. Adapt table and column names to match the database used in the course lab.

```
SELECT launch_site, COUNT(*) AS total_launches
FROM spacex_launches
GROUP BY launch_site
ORDER BY total_launches DESC;

SELECT orbit, COUNT(*) AS launches
FROM spacex_launches
GROUP BY orbit
ORDER BY launches DESC;

SELECT launch_year,
AVG(CASE WHEN landing_success=1 THEN 1.0 ELSE 0.0 END) AS success_rate
FROM spacex_launches
GROUP BY launch_year
ORDER BY launch_year;
```

Required result categories: launch sites, payloads, success rates, rankings, and time analysis. Include actual query outputs and the repository URL.

8. Interactive Visual Analytics

Folium map	Plotly Dash dashboard
• Center the map on the launch region. • Add markers using validated launch-site coordinates. • Add popups with site or launch details. • Include proximity analysis only when coordinates support it.	• Add interactive filters such as launch site, year, or orbit. • Pie chart: successful vs unsuccessful outcomes. • Scatter plot: payload mass vs launch outcome. • Include a screenshot and GitHub URL.

Evidence required: screenshot of the rendered map, screenshot of the running dashboard, and a short description of the interactivity. No map or dashboard was generated in this draft.

9. EDA Visualization Results

Visualization	Question answered	Actual finding
Scatter plot	How does payload mass relate to outcome?	Insert notebook result
Bar chart	Which sites have most launches/successful landings?	Insert notebook result
Yearly trend	How does launch activity/success rate change over time?	Insert notebook result
Distribution plot	What are payload characteristics?	Insert notebook result

This draft contains no computed chart results because no lab outputs were supplied.

10. EDA with SQL — Results

- Launch-site ranking: counts grouped by site.
- Payload analysis: average, median, minimum, and maximum where available.
- Success rate: successful outcomes divided by eligible attempts.
- Orbit ranking: launch counts and outcome rates by orbit.
- Time analysis: yearly launch volume and success rate.
- Check denominators and handle unknown outcomes transparently.

Replace this checklist with actual SQL result tables or screenshots from the completed lab.

11. Folium Map — Launch Sites and Proximity

Show launch-site markers at documented latitude and longitude. Popups can display the site name, location, and relevant launch records. Proximity analysis may compare sites with nearby geographic features if the required coordinates are available.

Workflow: Validate coordinates → create Folium map → add markers and popups → calculate distances only where supported.

Insert the actual map screenshot and coordinate source. No map was generated in this draft.

12. Plotly Dash — Interactive Results

The dashboard should allow exploration of launch outcomes using relevant filters. Keep labels clear and identify the dataset source.

- Pie chart: distribution of successful and unsuccessful outcomes.
- Scatter plot: payload mass versus landing outcome.
- Filters: launch site, year, or orbit, if supported by the data.
- Test that filters respond and include a screenshot plus repository URL.

Dashboard screenshots and measured values must be added from the working application.

13. Predictive Analysis — Model Comparison

Treat landing outcome as a binary classification task. Candidate models include Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors. Fit preprocessing only on training data, preferably in a scikit-learn pipeline.

Model	Purpose	Evaluation result
Logistic Regression	Interpretable baseline	Run model; enter metrics
Support Vector Machine	Margin-based comparison	Run model; enter metrics
Decision Tree	Rule-based model	Run model; enter metrics
K-Nearest Neighbors	Distance-based comparison	Run model; enter metrics

Report accuracy, precision, recall, F1-score, and a confusion matrix. Choose the best model using held-out evaluation. No model scores are claimed here.

14. Conclusion and Limitations

This project framework connects data collection, cleaning, descriptive analysis, geospatial visualization, dashboarding, and supervised learning to a practical launch-outcome question. The final conclusion must be based on completed analysis rather than assumptions.

- Limitations may include missing or ambiguous records, class imbalance, and changes in launch practices over time.
- A landing label may be unavailable for some historical records.
- Associations do not establish causation.
- Complete the labs, replace placeholders, test the dashboard and model pipeline, and add screenshots.

Final checklist: completed presentation PDF • GitHub URL • actual EDA charts • SQL results • Folium map • Plotly Dash evidence • model comparison and confusion matrix.

15. References and Reproducibility

- SpaceX API endpoint used in the course lab: record exact endpoint.
- Course-provided web-scraping source: record exact URL.
- Libraries as used: pandas, NumPy, Matplotlib, Seaborn, scikit-learn, SQL tools, Folium, Plotly, and Dash.
- GitHub repository: ADD YOUR REPOSITORY URL.
- Student: _____ Submission date: _____

Integrity note: This is a structured emergency draft, not evidence that the pipeline, charts, dashboard, or models were executed. Replace pending items with actual outputs if possible.